

LIFEWS AgriShine(TM) GrowMeal(TM)

JSS 1-3 SCIENTIFIC INVESTIGATION JOURNAL

STUDENT EDITION

SYSTEM / RESEARCH SKETCH

Name: _____

Class: _____ School: _____

**DEFINE -> HYPOTHESIZE -> EXPERIMENT -> QUANTIFY -> ANALYZE -> CONCLUDE ->
RECOMMEND**

SCIENTIFIC / ENGINEERING RECORD STANDARD

Use dated, traceable records. Define variables and units before collecting data. Record unexpected values rather than deleting them. Distinguish observations from interpretations. Follow teacher-approved safety procedures and instrument instructions.

Record item	What I must document

MY INSTRUMENT & DATA-QUALITY REGISTER

Instrument / method	Variable	Unit / range	Calibration / quality check

Where relevant, record sample size, replicate number, treatment, control, timestamp, sensor ID, uncertainty, missing data and corrective action.

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 01 - Production Garden

QUESTION / OBJECTIVE: How efficiently is the production area established?

METHOD: Measure planted and established populations in sample rows.

PRIMARY METRIC: Establishment rate (%)

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Calculate establishment rate and identify causes of loss.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 02 - Production Garden

QUESTION / OBJECTIVE: How does crop growth change with time?

METHOD: Measure the same sample plants on repeated dates.

PRIMARY METRIC: Growth rate

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Calculate change per unit time.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 03 - Production Garden

QUESTION / OBJECTIVE: How uniform is the crop stand?

METHOD: Measure a response across several locations.

PRIMARY METRIC: Mean, range, variation

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify spatial variability.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 04 - Production Garden

QUESTION / OBJECTIVE: What inputs are used to produce the harvest?

METHOD: Record selected water, labor or material inputs.

PRIMARY METRIC: Input per output

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Relate input use to production.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 05 - Production Garden

QUESTION / OBJECTIVE: What limits production performance?

METHOD: Combine growth, loss and harvest records.

PRIMARY METRIC: Yield and loss indicators

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify the dominant constraint and recommend action.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

PRODUCTION GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 06 - Raised Systems Garden

QUESTION / OBJECTIVE: How does root-zone condition vary within the bed?

METHOD: Sample moisture or another approved indicator at several positions.

PRIMARY METRIC: Spatial root-zone metric

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Compare edge and center conditions.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 07 - Raised Systems Garden

QUESTION / OBJECTIVE: What is the bed's effective growing volume?

METHOD: Measure bed dimensions and estimate volume.

PRIMARY METRIC: Volume

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: State geometric assumptions.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 08 - Raised Systems Garden

QUESTION / OBJECTIVE: How quickly does water infiltrate?

METHOD: Run repeated teacher-approved infiltration observations.

PRIMARY METRIC: Infiltration rate

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Compare locations and variability.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 09 - Raised Systems Garden

QUESTION / OBJECTIVE: How effective is drainage?

METHOD: Observe drainage response after a controlled watering event.

PRIMARY METRIC: Drainage time / score

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify possible restrictions.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 10 - Raised Systems Garden

QUESTION / OBJECTIVE: Does the raised system use resources efficiently?

METHOD: Relate output to bed area or selected input.

PRIMARY METRIC: Productivity metric

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Evaluate one advantage and one limitation.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

RAISED SYSTEMS GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 11 - Grid Experiment Garden

QUESTION / OBJECTIVE: Does treatment affect the response variable?

METHOD: Use treatment, control and replication in labeled grid cells.

PRIMARY METRIC: Treatment effect

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Compare group means or typical responses.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 12 - Grid Experiment Garden

QUESTION / OBJECTIVE: Is replication adequate?

METHOD: Inspect within-treatment variability.

PRIMARY METRIC: Range / spread

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Explain why replication matters.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 13 - Grid Experiment Garden

QUESTION / OBJECTIVE: Did randomization reduce positional bias?

METHOD: Map treatment locations and responses.

PRIMARY METRIC: Position vs response

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Look for location patterns.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 14 - Grid Experiment Garden

QUESTION / OBJECTIVE: Are controlled variables consistent?

METHOD: Audit agreed constants during the trial.

PRIMARY METRIC: Control checklist

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify protocol deviations.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 15 - Grid Experiment Garden

QUESTION / OBJECTIVE: What conclusion is supported?

METHOD: Summarize treatment and control evidence.

PRIMARY METRIC: Effect direction/magnitude

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: State conclusion and limitations.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

GRID EXPERIMENT GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 16 - Crop Rotation Garden

QUESTION / OBJECTIVE: How diverse is the rotation?

METHOD: Classify crop families across the sequence.

PRIMARY METRIC: Family diversity

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Evaluate repetition.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 17 - Crop Rotation Garden

QUESTION / OBJECTIVE: What is the return interval for each family?

METHOD: Trace bed-history records.

PRIMARY METRIC: Rotation interval

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify short return intervals.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 18 - Crop Rotation Garden

QUESTION / OBJECTIVE: Where should legumes enter the sequence?

METHOD: Compare two rotation plans.

PRIMARY METRIC: Sequence performance

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Justify placement.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 19 - Crop Rotation Garden

QUESTION / OBJECTIVE: How does cover vary between phases?

METHOD: Estimate soil cover after selected crops.

PRIMARY METRIC: Cover (%) / category

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Relate cover to soil protection.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 20 - Crop Rotation Garden

QUESTION / OBJECTIVE: Which rotation best meets production goals?

METHOD: Score alternative sequences against criteria.

PRIMARY METRIC: Decision score

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Recommend a rotation.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

CROP ROTATION GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 21 - Intercropping Garden

QUESTION / OBJECTIVE: How are component crops distributed?

METHOD: Measure row or plant densities for both crops.

PRIMARY METRIC: Component density

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Check intended proportions.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 22 - Intercropping Garden

QUESTION / OBJECTIVE: Do canopy layers differ?

METHOD: Measure height or canopy position.

PRIMARY METRIC: Canopy stratification

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret complementarity/competition.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 23 - Intercropping Garden

QUESTION / OBJECTIVE: How do component yields compare?

METHOD: Measure harvest from defined sample areas.

PRIMARY METRIC: Component yield

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Compare monocrop-reference values if available.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 24 - Intercropping Garden

QUESTION / OBJECTIVE: What is the land equivalent ratio?

METHOD: Use component intercrop and reference yields.

PRIMARY METRIC: LER

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret LER above, equal to, or below 1.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 25 - Intercropping Garden

QUESTION / OBJECTIVE: What trade-off does the intercrop create?

METHOD: Compare output, management and competition indicators.

PRIMARY METRIC: Multiple criteria

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Evaluate the system.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INTERCROPPING GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 26 - Water-Smart Garden

QUESTION / OBJECTIVE: What is root-zone moisture status?

METHOD: Measure approved moisture indicators at representative points.

PRIMARY METRIC: Soil moisture

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Assess variability.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 27 - Water-Smart Garden

QUESTION / OBJECTIVE: How much irrigation is required?

METHOD: Use area, target depth/volume and current condition.

PRIMARY METRIC: Irrigation requirement

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Show assumptions.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 28 - Water-Smart Garden

QUESTION / OBJECTIVE: How uniform is application?

METHOD: Collect or estimate water at several points.

PRIMARY METRIC: Distribution uniformity

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify weak zones.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 29 - Water-Smart Garden

QUESTION / OBJECTIVE: How efficient is irrigation?

METHOD: Compare useful application with supplied water where measurable.

PRIMARY METRIC: Efficiency (%)

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify losses.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 30 - Water-Smart Garden

QUESTION / OBJECTIVE: What is water productivity?

METHOD: Relate harvest to measured water input.

PRIMARY METRIC: Yield per water

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret the indicator.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

WATER-SMART GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 31 - Compost and Soil Garden

QUESTION / OBJECTIVE: How does pile temperature evolve?

METHOD: Measure at consistent depth and time.

PRIMARY METRIC: Temperature time series

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret decomposition phase.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 32 - Compost and Soil Garden

QUESTION / OBJECTIVE: Is moisture suitable for decomposition?

METHOD: Use teacher-approved moisture assessment.

PRIMARY METRIC: Moisture category

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Recommend adjustment if needed.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 33 - Compost and Soil Garden

QUESTION / OBJECTIVE: How much does pile volume decline?

METHOD: Measure dimensions on repeated dates.

PRIMARY METRIC: Volume change (%)

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret decomposition progress.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 34 - Compost and Soil Garden

QUESTION / OBJECTIVE: How mature is the compost?

METHOD: Use agreed physical maturity indicators.

PRIMARY METRIC: Maturity score

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: State evidence and uncertainty.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 35 - Compost and Soil Garden

QUESTION / OBJECTIVE: How does amended soil differ?

METHOD: Compare approved soil indicators with and without mature compost.

PRIMARY METRIC: Soil response metric

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret the treatment effect.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

COMPOST AND SOIL GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 36 - Soilless Learning Garden

QUESTION / OBJECTIVE: How is solution delivered through the system?

METHOD: Map reservoir, delivery, root zone and return/drainage.

PRIMARY METRIC: Flow architecture

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Identify failure points.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 37 - Soilless Learning Garden

QUESTION / OBJECTIVE: How stable is pH?

METHOD: Measure pH at consistent times.

PRIMARY METRIC: pH time series

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret drift.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 38 - Soilless Learning Garden

QUESTION / OBJECTIVE: How stable is electrical conductivity?

METHOD: Measure EC with supervision.

PRIMARY METRIC: EC time series

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Interpret dissolved-ion concentration changes.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 39 - Soilless Learning Garden

QUESTION / OBJECTIVE: Is flow adequate?

METHOD: Measure volume delivered over time.

PRIMARY METRIC: Flow rate

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Compare locations.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 40 - Soilless Learning Garden

QUESTION / OBJECTIVE: What evidence indicates root-zone stress?

METHOD: Combine root, solution and plant observations.

PRIMARY METRIC: Diagnostic indicators

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Propose corrective action.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

SOILLESS LEARNING GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 41 - Sensor-Ready Garden

QUESTION / OBJECTIVE: What exactly does the sensor measure?

METHOD: Document measurand, unit, range and location.

PRIMARY METRIC: Sensor specification

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Check fitness for purpose.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 42 - Sensor-Ready Garden

QUESTION / OBJECTIVE: Is the sensor repeatable?

METHOD: Take repeated readings under stable conditions.

PRIMARY METRIC: Spread / repeatability

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Evaluate precision.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 43 - Sensor-Ready Garden

QUESTION / OBJECTIVE: Does it agree with a reference?

METHOD: Compare sensor and teacher-approved reference readings.

PRIMARY METRIC: Error

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Assess accuracy.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 44 - Sensor-Ready Garden

QUESTION / OBJECTIVE: Is placement representative?

METHOD: Compare readings at candidate locations.

PRIMARY METRIC: Spatial difference

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Justify placement.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 45 - Sensor-Ready Garden

QUESTION / OBJECTIVE: Are there outliers or drift?

METHOD: Inspect a short time series.

PRIMARY METRIC: Outlier/drift evidence

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Flag suspect data and next check.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

SENSOR-READY GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

SYSTEM / EXPERIMENT MAP

Map the system boundary, components, flows, treatment areas, sensors, sampling points or control elements relevant to this garden.

MAP / SCHEMATIC / EXPERIMENTAL LAYOUT

Primary question or design objective: _____

INVESTIGATION 46 - Applied Systems Garden

QUESTION / OBJECTIVE: What is inside the system boundary?

METHOD: Map components, inputs, outputs and environment.

PRIMARY METRIC: System boundary

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Explain inclusions/exclusions.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 47 - Applied Systems Garden

QUESTION / OBJECTIVE: What are the main stocks and flows?

METHOD: Identify stored quantities and transfer pathways.

PRIMARY METRIC: Stock/flow map

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Trace one pathway.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 48 - Applied Systems Garden

QUESTION / OBJECTIVE: Where is a feedback loop?

METHOD: Map a response that influences a future input or state.

PRIMARY METRIC: Feedback structure

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Explain reinforcing/balancing behavior.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 49 - Applied Systems Garden

QUESTION / OBJECTIVE: What trade-off exists?

METHOD: Compare two objectives using evidence.

PRIMARY METRIC: Trade-off metrics

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Explain the compromise.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

INVESTIGATION 50 - Applied Systems Garden

QUESTION / OBJECTIVE: Which scenario performs best?

METHOD: Compare two management scenarios against criteria.

PRIMARY METRIC: Scenario score

Replicate / time	Condition / input	Measurement	Unit / QC note

CALCULATION / GRAPH / MODEL / SCHEMATIC

ANALYSIS: Recommend a decision.

Result / conclusion: _____

Uncertainty / limitation: _____

Recommendation / next test: _____

Date: _____ Group: _____ Instrument/Sensor ID: _____ Teacher: _____

APPLIED SYSTEMS GARDEN - EVIDENCE REVIEW

BEST GRAPH / RESULT / DESIGN EVIDENCE

Most defensible finding: _____

Main limitation: _____

Management, research or design implication: _____

INDEPENDENT CAPSTONE INVESTIGATION

Research/design question: _____

Hypothesis or performance target: _____

Independent/design variable(s): _____

Response/performance variable(s): _____

Controls, constraints and assumptions: _____

Run / replicate	Input / treatment	Response	Unit / QC

ANALYSIS / MODEL / GRAPH

Evidence-based conclusion and recommendation: _____

MY GROWMEAL(TM) JSS 1-3 CERTIFICATE

This journal belongs to

who completed the GrowMeal(TM) investigation and research journal.

**DEFINE -> HYPOTHESIZE -> EXPERIMENT -> QUANTIFY -> ANALYZE -> CONCLUDE ->
RECOMMEND**

Teacher: _____ Date: _____